Welcome to LUG!



Please check in: sh.lug.umbc.edu/checkin

OpenStreetMap

What is OpenStreetMap?



vs.



- Not just a “Google Maps Alternative”
  - OSM is a Free, editable crowdsourced spatial database of the entire world.

What is OpenStreetMap

Sometimes considered the “Wikipedia” of google maps

OSM is the truly **OPEN** map platform

What makes OSM different from other map platforms

- No locked down / restricted data
- No API fees
- No restrictions of offline data caching
- User location tracking



Licensing

OSM is licensed under the **Open Database License** (ODbL) created by the Open Data Commons

You are free:

- *To share*: To copy, distribute and use the database.
- *To create*: To produce works from the database.
- *To adapt*: To modify, transform and build upon the database.

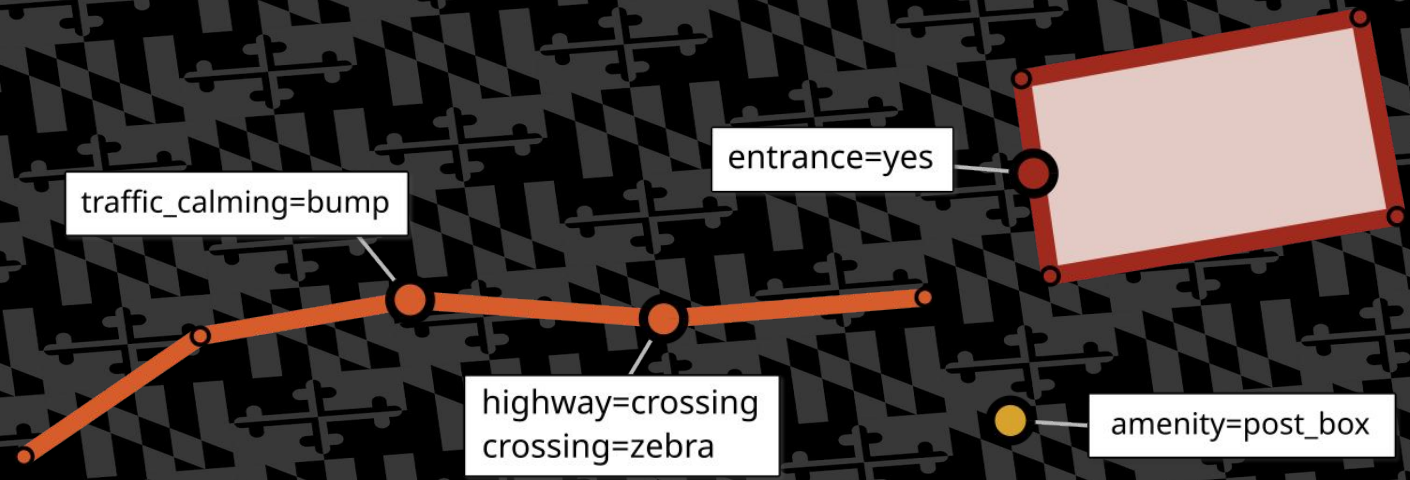
As long as you:

- *Attribute*: You must attribute any public use of the database, or works produced from the database, in the manner specified in the ODbL. For any use or redistribution of the database, or works produced from it, you must make clear to others the license of the database and keep intact any notices on the original database.
- *Share-Alike*: If you publicly use any adapted version of this database, or works produced from an adapted database, you must also offer that adapted database under the ODbL.
- *Keep open*: If you redistribute the database, or an adapted version of it, then you may use technological measures that restrict the work (such as DRM) as long as you also redistribute a version without such measures.

How OSM Works

Made up of two primary elements: **nodes** and **ways**

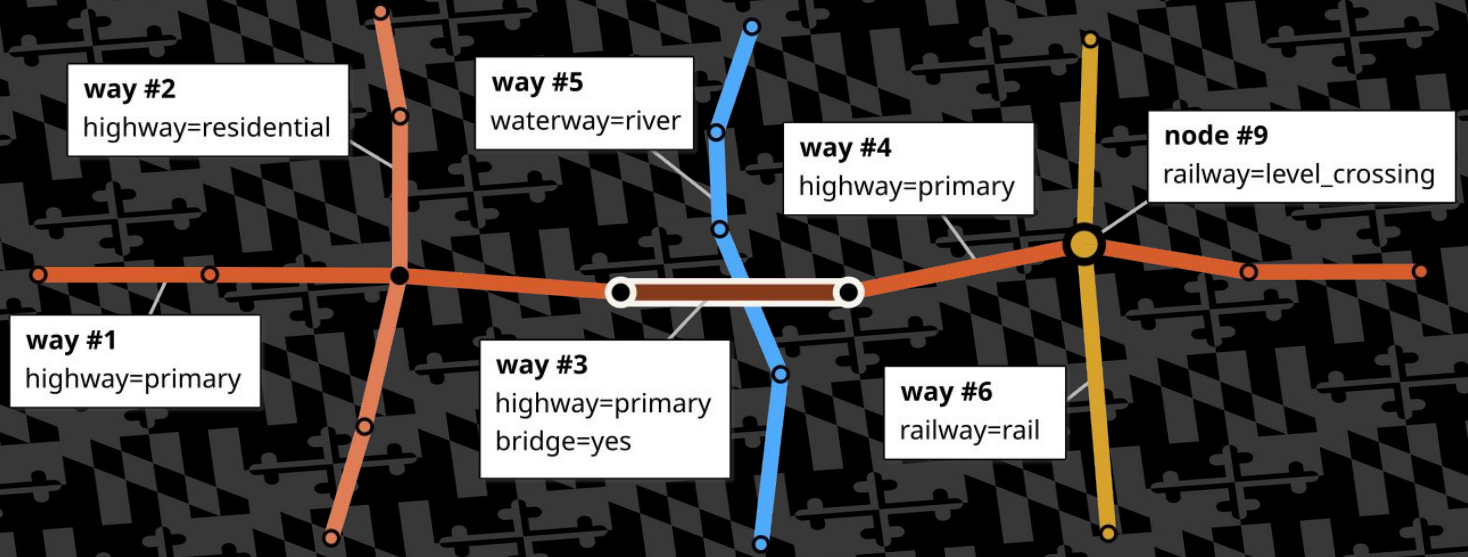
- A way is an ordered list of nodes
- A way that is closed can form a polygon



How OSM Works

OSM elements can have one or more **tags**

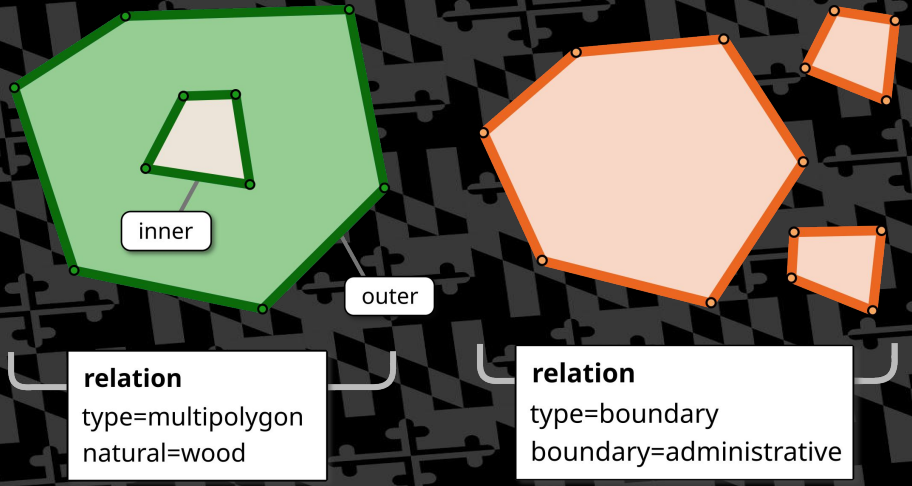
- Attributes that describe what kind of feature or element is referenced

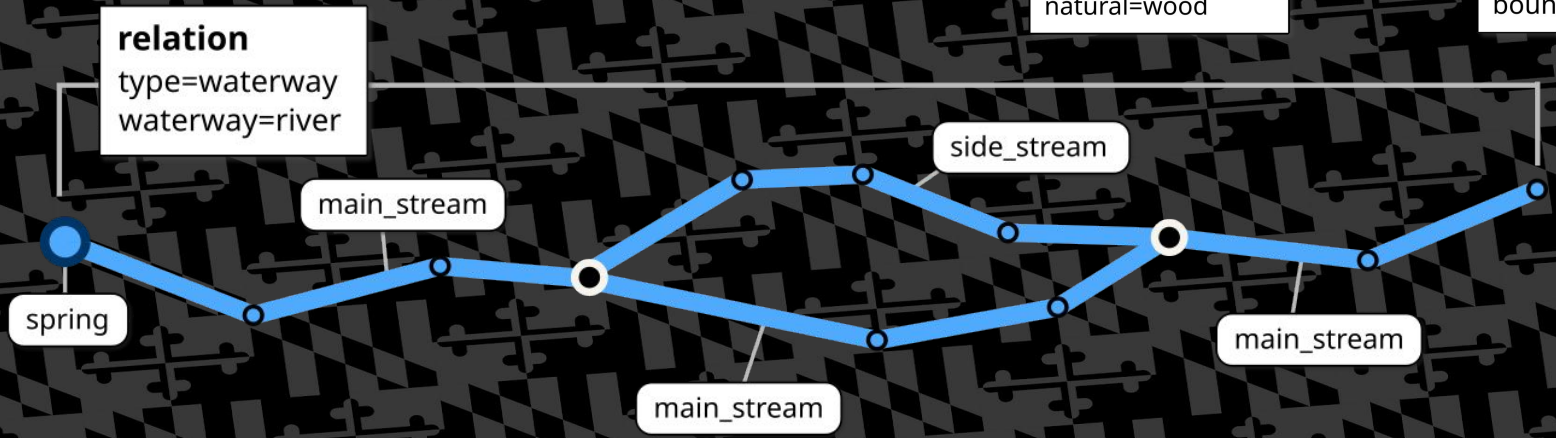


How OSM Works

The third element: **relations**

Used to combine multiple elements into more complex structures



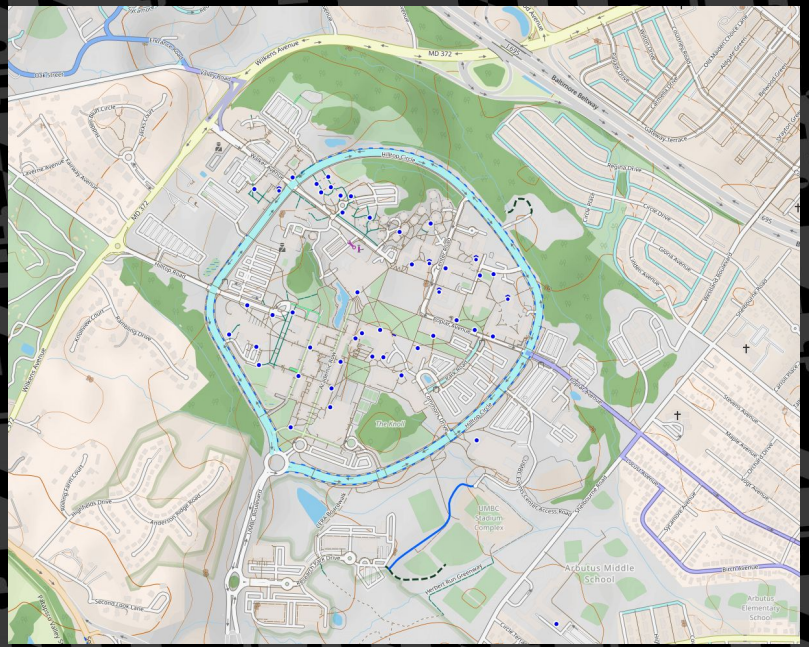


Displaying and using the data

OSM data is stored as vectors. Not pre-rendered.

Can be rendered in any manner you like

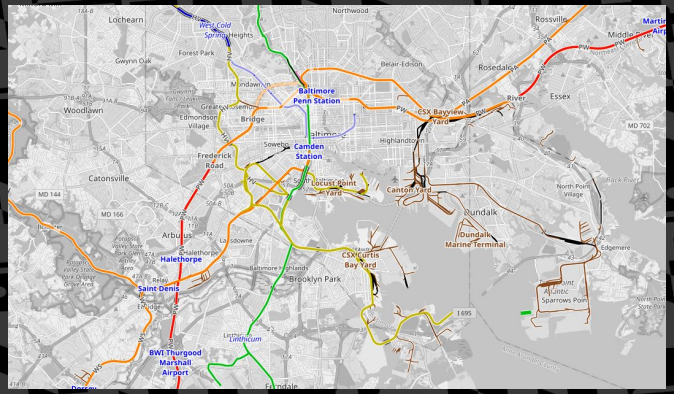


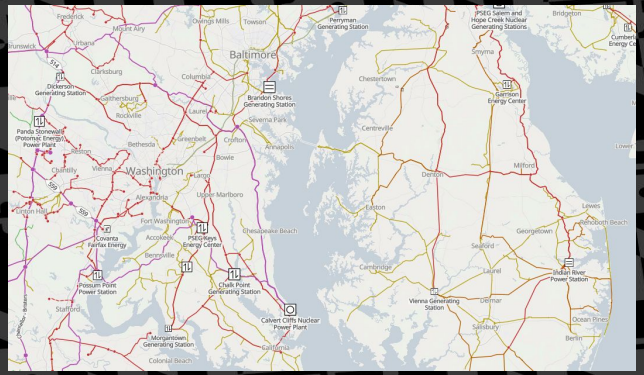


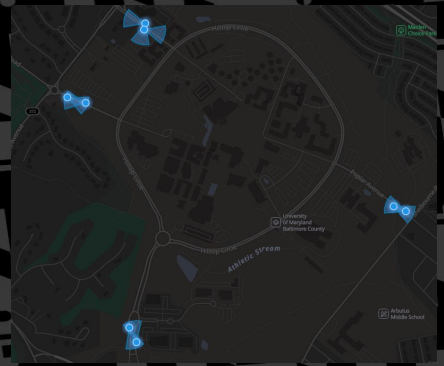
OSM in other applications

All representations of OSM data

- OpenRailwayMap
- OpenTopoMap
- Open Infrastructure Map
- DeFlock
- Open Source Routing Machine







Using OSM

You can use OSM data in your own project too.

So many possibilities for both rendering and analysis.

You can download the planet for just **89GB**

Lots of tools and libraries

- OverPass API / Overpass Turbo
- OSM2pgsql - Convert to PostgresQL
- Osmium - C++ framework
- Osm4j - Java framework

Contributing to OSM

Anyone can contribute to OSM!

Create an account and use the web editor (iD)

JOSM - more advanced

Mobile apps for fieldmapping

- Every Door
- Vespucci
- StreetComplete

Ways of contributing

- Find missing things; Look at the map, use mapcomplete.org
- Micromapping; map small features in an area
- Look at mapping projects (Some broad some local)
  https://wiki.openstreetmap.org/wiki/Mapping_projects
- 31 Day OSM Challenge
  https://wiki.openstreetmap.org/wiki/31_Day_OSM_Challenge

The impact of OSM

- Foss mobile apps and applications that need a map
- Used by big names in the Tech Industry
- Used for logistics & Research, Urban Planning, Transport routing, GIS Research

OSM by the numbers:

- 10 million registered users
- 10 billion nodes
- 4 million map changes/day

Things left to mention

Contributing (how-to). Compare UMD to UMBC

Micro-mapping

Ways to learn contributing

Ways to figure out what to contribute

Why osm matters